

SUMMARY STATISTICS

new 1.7–1.8 • old 1.7–1.8

ZONE A · THE FORMULAS

$$\bar{x} = \frac{\sum x_i}{n}$$

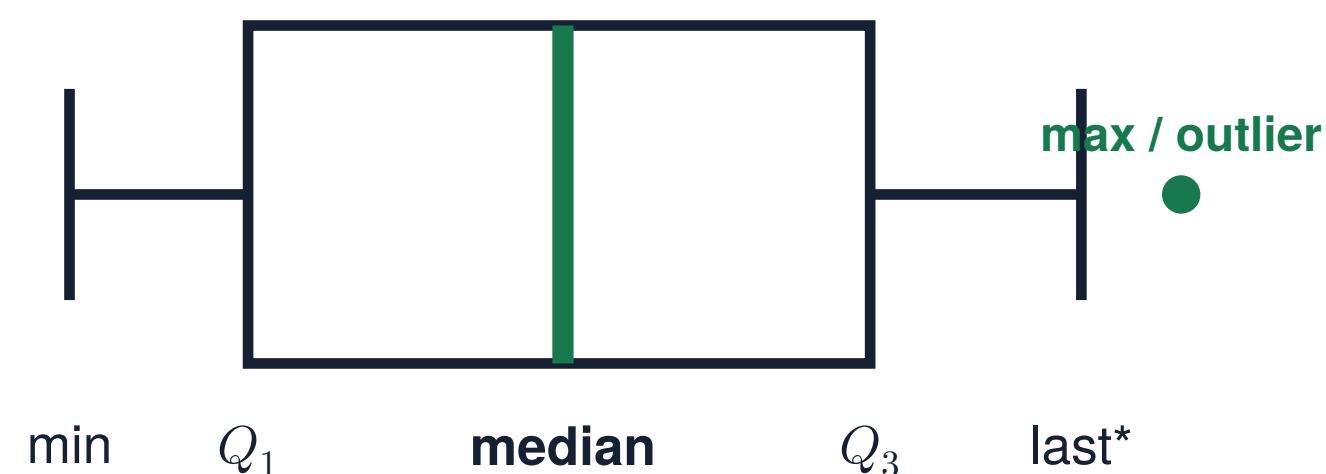
$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

$$s_x = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}}$$

$$\text{IQR} = Q_3 - Q_1$$

outlier if $x > Q_3 + 1.5(\text{IQR})$
or $x < Q_1 - 1.5(\text{IQR})$

A BOX PLOT IS A FIVE-NUMBER STORY



*The whisker stops at the last non-outlier. The maximum here is the separate point.

CENTER + SPREAD WORK AS A PAIR

symmetric, no outliers $\rightarrow \bar{x}$ and s
skewed or outliers $\rightarrow$ median and IQR

ZONE B · WHAT THE SYMBOLS MEAN

SAMPLE

$\bar{x}$ = sample mean s = sample SD
five-number summary: min, Q_1 , median, Q_3 , max

POPULATION + RESISTANCE

μ = population mean	σ = population SD
resistant: median, IQR	not resistant: mean, SD,
range	

ZONE C · SAY IT IN WORDS

“Typically, _____ varies from the mean by about _____.”

variable in context

value of s

units

ZONE D · WATCH OUT

THE DENOMINATOR

For a **sample** variance or SD, divide by $n - 1$, not n .

TRANSFORMATIONS

Adding a constant shifts the center but not the spread. Multiplying by a constant scales **both**.